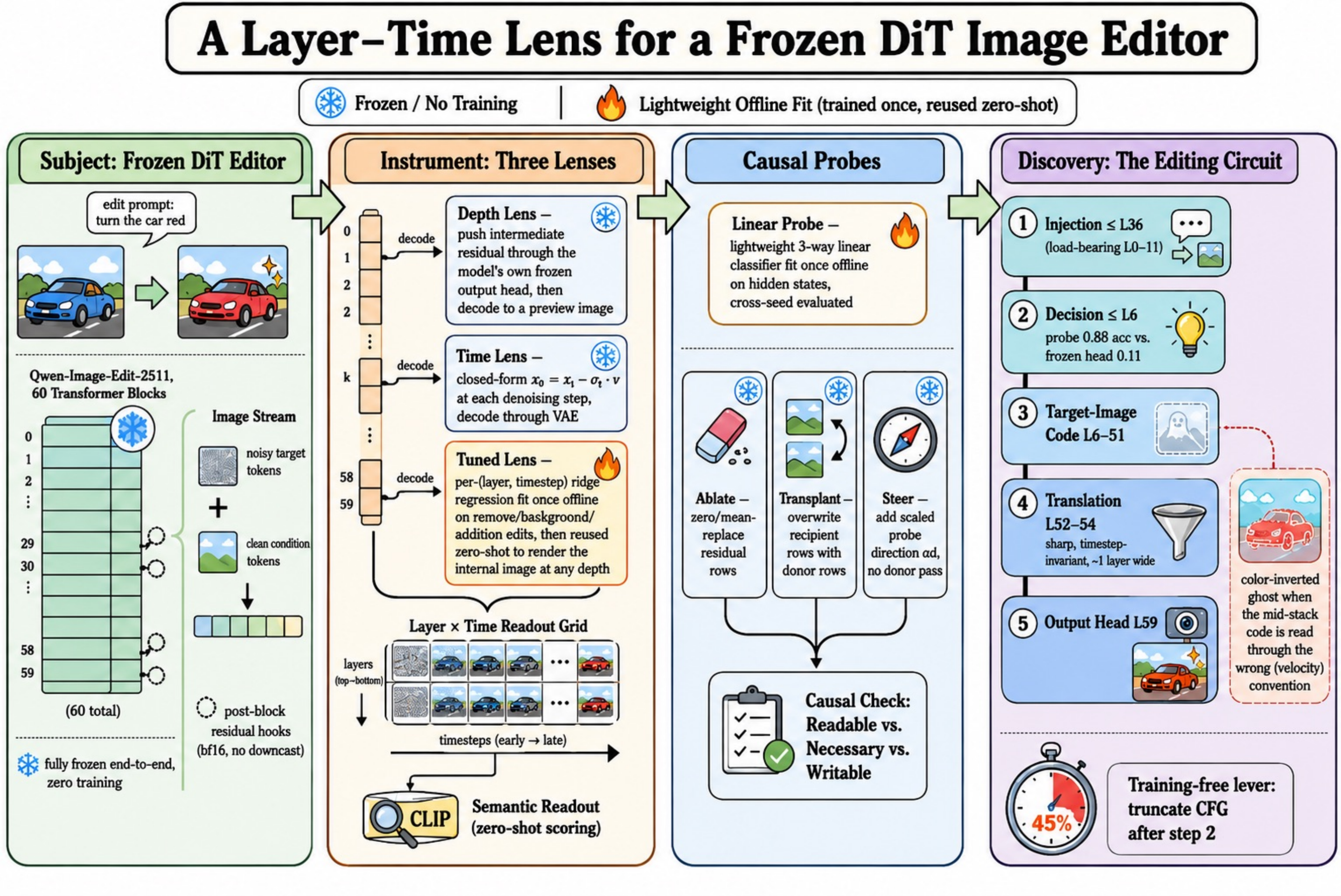

A Layer–Time Lens for a Frozen DiT Image Editor
Frozen / No Training
Lightweight Offline Fit (trained once, reused zero-shot)
Subject: Frozen DiT Editor
edit prompt: turn the car red
Qwen-Image-Edit-2511, 60 Transformer Blocks
0
1
2
29
30
58
59
(60 total)
Image Stream
noisy target tokens
+
clean condition tokens
post-block residual hooks (bf16, no downcast)
fully frozen end-to-end, zero training
Instrument: Three Lenses
0
1
2
2
k
58
59
decode
decode
decode
Depth Lens – push intermediate residual through the model's own frozen output head, then decode to a preview image
Time Lens – closed-form $x_0 = x_1 - \sigma_t \cdot v$ at each denoising step, decode through VAE
Tuned Lens – per-(layer, timestep) ridge regression fit once offline on remove/baskground/addition edits, then reused zero-shot to render the internal image at any depth
Layer × Time Readout Grid
layers (top→bottom)
timesteps (early → late)
CLIP
Semantic Readout (zero-shot scoring)
Causal Probes
Linear Probe – lightweight 3-way linear classifier fit once offline on hidden states, cross-seed evaluated
Ablate – zero/mean-replace residual rows
Transplant – overwrite recipient rows with donor rows
Steer – add scaled probe direction αd, no donor pass
Causal Check: Readable vs. Necessary vs. Writable
Discovery: The Editing Circuit
1 Injection ≤ L36 (load-bearing L0–11)
2 Decision ≤ L6 probe 0.88 acc vs. frozen head 0.11
3 Target-Image Code L6–51
4 Translation L52–54 sharp, timestep-invariant, ~1 layer wide
5 Output Head L59
color-inverted ghost when the mid-stack code is read through the wrong (velocity) convention
45%
Training-free lever: truncate CFG after step 2